1) a)

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

So $\lim_{x\to 0}\left[\frac{\ln(1+x)}{x}\right] = \lim_{x\to 0}\left[\frac{x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots}{x}\right] = \lim_{x\to 0}\left[1 - \frac{x}{2} + \frac{x^2}{3} - \cdots\right] = 1$

1) b)

$e^x = 1 + x + \frac{x^2}{2} + \cdots$

So $\lim_{x\to 0}\left[\frac{e^x-1}{2x}\right] = \lim_{x\to 0}\left[\frac{x+\frac{x^2}{2}+\cdots}{2x}\right] = \lim_{x\to 0}\left[\frac{1}{2} + \frac{x}{4} + \cdots\right] = \frac{1}{2}$

1) c)

$\cos x + \sin x = 1 - \frac{1}{2}x^2 + \cdots + x - \frac{1}{6}x^3 + \cdots$

$e^x = 1 + x + \frac{1}{2}x^2 + \cdots$

So $\lim_{x\to 0}\left[\frac{\cos x+\sin x}{e^x}\right] = \lim_{x\to 0}\left[\frac{1+x-\frac{1}{2}x^2-\frac{1}{6}x^3+\cdots}{1+x+\frac{1}{2}x^2}\right] = 1$

2) a)

$\ln(1+x^2) = (x^2) - \frac{(x^2)^2}{2} + \frac{(x^2)^3}{3} - \cdots = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \cdots$

So $\lim_{x\to 0}\left[\frac{\ln(1+x^2)}{x^2}\right] = \lim_{x\to 0}\left[\frac{x^2-\frac{x^4}{2}+\frac{x^6}{3}-\cdots}{x^2}\right] = \lim_{x\to 0}\left[1 - \frac{x^2}{2} + \frac{x^4}{3} - \cdots\right] = 1$

2) b)

$(1+3x)^{-2} = 1 - 2\times 3x + \frac{-2\times -3}{2}(3x)^2 - \cdots = 1 - 6x + 27x^2 - \cdots$

So $\lim_{x\to 0}\left[\frac{-1+(1+3x)^{-2}}{x}\right] = \lim_{x\to 0}\left[\frac{-1+1-6x+27x^2-\cdots}{x}\right] = \lim_{x\to 0}[-6 + 27x - \cdots] = -6$

2) c)

$\sin 4x = 4x - \frac{1}{6}(4x)^3 + \cdots = 4x - \frac{32}{2}x^3 + \cdots$

So $\lim_{x\to 0}\left[\frac{\sin 4x}{x}\right] = \lim_{x\to 0}\left[\frac{4x-\frac{32}{2}x^3+\cdots}{x}\right] = \lim_{x\to 0}\left[4 - \frac{32}{2}x^2 + \cdots\right] = 4$

2) d)

$\cos 3x = 1 - \frac{1}{2}(3x)^2 + \cdots = 1 - \frac{9}{2}x^2 + \cdots$

So $\lim_{x\to 0}\left[\frac{x^2}{1-\cos 3x}\right] = \lim_{x\to 0}\left[\frac{x^2}{1-\left(1-\frac{9}{2}x^2+\cdots\right)}\right] = \lim_{x\to 0}\left[\frac{x^2}{\frac{9}{2}x^2+\cdots}\right] \approx \lim_{x\to 0}\left[\frac{1}{\frac{9}{2}}\right] = \frac{1}{\frac{9}{2}} = \frac{2}{9}$

2) e)

$e^{-2x} = 1 - 2x + \frac{1}{2}(-2x)^2 + \cdots = 1 - 2x + 2x^2 + \cdots$

So $\lim_{x\to 0}\left[\frac{1-e^{-2x}}{x}\right] = \lim_{x\to 0}\left[\frac{1-(1-2x+2x^2+\cdots)}{x}\right] = \lim_{x\to 0}\left[\frac{2x-2x^2+\cdots}{x}\right] \approx \lim_{x\to 0}[2-2x] = 2$

2) f)

$e^{3x} = 1 + 3x + \frac{1}{2}(3x)^2 + \cdots = 1 + 3x + \frac{9}{2}x^2 + \cdots$

$e^{x} = 1 + x + \frac{1}{2}x^2 + \cdots$

So $\lim_{x\to 0}\left[\frac{e^{3x}-e^{x}}{x}\right] = \lim_{x\to 0}\left[\frac{1+3x+\frac{9}{2}x^2+\cdots-\left(1+x+\frac{1}{2}x^2+\cdots\right)}{x}\right] = \lim_{x\to 0}\left[\frac{2x+4x^2+\cdots}{x}\right]$

$\approx \lim_{x\to 0}[2+4x] = 2$

2) g)

$e^{2x} = 1 + 2x + \frac{1}{2}(2x)^2 + \cdots = 1 + 2x + 2x^2 + \cdots$

So $\lim_{x\to 0}\left[\frac{e^{2x}-2}{2e^{2x}+1}\right] = \lim_{x\to 0}\left[\frac{1+2x+2x^2+\cdots-2}{2(1+2x+2x^2+\cdots)+1}\right] \approx \lim_{x\to 0}\left[\frac{-1+2x+2x^2}{3+2x+4x^2}\right] = -\frac{1}{3}$

3)

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

$\cos x = 1 - \frac{1}{2}x^2 + \frac{1}{4}x^4 \ldots$

So $\lim_{x\to 0}\left[\frac{\ln(1+x)-x}{1-\cos x}\right] = \lim_{x\to 0}\left[\frac{x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots-x}{1-1+\frac{1}{2}x^2-\frac{1}{4}x^4+\cdots}\right] = \lim_{x\to 0}\left[\frac{-\frac{x^2}{2}+\frac{x^3}{3}-\cdots}{\frac{1}{2}x^2-\frac{1}{4}x^4+\cdots}\right]$

$= \lim_{x\to 0}\left[\frac{-\frac{1}{2}+\frac{x}{3}-\cdots}{\frac{1}{2}x^2-\frac{1}{4}x^4+\cdots}\right] \approx \lim_{x\to 0}\left[\frac{-\frac{1}{2}+\frac{x}{3}}{\frac{1}{2}-\frac{1}{4}x^2}\right] = -1$

4) a)

$$\sqrt{x^2+5x} = x\sqrt{1+\frac{5}{x}} = x\left(1+\frac{5}{x}\right)^{\frac{1}{2}} = x\left[1+\frac{5}{2x}-\frac{25}{8x^2}+\cdots\right] = x+\frac{5}{2}-\frac{25}{8x}+\cdots$$

$$\text{So } \lim_{x\to\infty}\left[x-\sqrt{x^2+5x}\right] = \lim_{x\to\infty}\left[x-\left(x+\frac{5}{2}-\frac{25}{8x}+\cdots\right)\right] \approx \lim_{x\to\infty}\left[-\frac{5}{2}+\frac{25}{8x}\right] = -\frac{5}{2}$$

4) b)

$$\sqrt[3]{x^3+7} = x\left(1+\frac{7}{x^3}\right)^{\frac{1}{3}} = x\left[1+\frac{7}{3x^3}-\frac{49}{9x^6}+\cdots\right] = x+\frac{7}{3x^2}-\frac{49}{9x^5}+\cdots$$

$$\text{So } \lim_{x\to\infty}\left[\sqrt[3]{x^3+7}-x\right] = \lim_{x\to\infty}\left[\left(x+\frac{7}{3x^2}-\frac{49}{9x^5}+\cdots\right)-x\right] \approx \lim_{x\to\infty}\left[\frac{7}{3x^2}-\frac{49}{9x^5}\right] = 0$$

4) c)

$$\sqrt{x^2-8x+3} = x\sqrt{1-\frac{8}{x}+\frac{3}{x^2}} = x\left(1+\left(-\frac{8}{x}+\frac{3}{x^2}\right)\right)^{\frac{1}{2}}$$

$$= x\left[1+\frac{1}{2}\left(-\frac{8}{x}+\frac{3}{x^2}\right)-\frac{\frac{1}{2}\times-\frac{1}{2}}{2}\left(-\frac{8}{x}+\frac{3}{x^2}\right)^2+\cdots\right]$$

$$= x\left[1-\frac{4}{x}+\frac{3}{2x^2}-\frac{9}{8x^4}+\frac{6}{x^3}-\frac{8}{x^2}+\cdots\right] = x-4-\frac{13}{2x}+\cdots$$

$$\text{So } \lim_{x\to\infty}\left[x-\sqrt{x^2-8x+3}\right] = \lim_{x\to\infty}\left[x-\left(x-4-\frac{13}{2x}+\cdots\right)\right] \approx \lim_{x\to\infty}\left[4+\frac{23}{2x}\right] = 4$$

5) a)

$$2x\cos 3x+\sin 2x = 2x\left(1-\frac{1}{2}(3x)^2+\frac{1}{24}(3x)^4+\cdots\right)+2x-\frac{1}{6}(2x)^3+\frac{1}{120}(2x)^5+\cdots$$

$$\approx 2x-9x^3+\frac{27}{4}x^5+2x-\frac{4}{3}x^3+\frac{4}{15}x^5$$

$$= 4x-\frac{31}{3}x^3+\frac{421}{60}x^5 \text{ as req.}$$

5) b)

$$\text{So } \lim_{x\to 0}\left[\frac{-4x+2x\cos 3x+\sin 2x}{x^3}\right] \approx \lim_{x\to 0}\left[\frac{-4x+\left(4x-\frac{31}{3}x^3+\frac{421}{60}x^5\right)}{x^3}\right] = \lim_{x\to 0}\left[\frac{-\frac{31}{3}x^3+\frac{421}{60}x^5}{x^3}\right]$$

$$= \lim_{x\to 0}\left[-\frac{31}{3}+\frac{421}{60}x^2\right] = -\frac{31}{3}$$

6) a)

$$\sin 3x = 3x-\frac{1}{6}(3x)^3+\frac{1}{120}(3x)^5+\cdots \approx 3x-\frac{9}{2}x^3+\frac{81}{40}x^5$$

6) b)

$\cos 4x = 1 - \frac{1}{2}(4x)^2 + \frac{1}{24}(4x)^4 + \cdots \approx 1 - 8x^2 + \frac{32}{3}x^4$

6) c)

$\lim_{x\to 0}\left[\frac{3x\cos 4x - \sin 3x}{2x^3}\right] \approx \lim_{x\to 0}\left[\frac{3x\left(1-8x^2+\frac{32}{3}x^4\right)-\left(3x-\frac{9}{2}x^3+\frac{81}{40}x^5\right)}{2x^3}\right]$

$= \lim_{x\to 0}\left[\frac{3x-24x^3+32x^5-3x+\frac{9}{2}x^3-\frac{81}{40}x^5}{2x^3}\right] = \lim_{x\to 0}\left[\frac{-\frac{39}{2}x^3+\frac{1199}{40}x^5}{2x^3}\right] = \lim_{x\to 0}\left[-\frac{39}{4}+\frac{1199}{80}x^2\right] = -\frac{39}{4}$

7) a)

$(1+3x)^{-\frac{1}{3}} = 1 - \frac{1}{3}(3x) + \frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)}{2}(3x)^2 + \frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)\left(-\frac{7}{3}\right)}{6}(3x)^3 \ldots \approx 1 - x + 2x^2 - \frac{14}{3}x^3$

So that $e^x(1+3x)^{-\frac{1}{3}} \approx \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3\right)\left(1 - x + 2x^2 - \frac{14}{3}x^3\right)$

$= 1 - x + 2x^2 - \frac{14}{3}x^3 + x - x^2 + 2x^3 - \frac{14}{3}x^4 + \frac{1}{2}x^2 - \frac{1}{2}x^3 + x^4 - \frac{7}{3}x^5 + \frac{1}{6}x^3 - \frac{1}{6}x^4 + \frac{1}{3}x^5 - \cdots$

$\approx 1 + \frac{3}{2}x^2 - 3x^3$ as req.

7) b)

So $\lim_{x\to 0}\left[\frac{1-f(x)}{xg(x)}\right] \approx \lim_{x\to 0}\left[\frac{1-\left(1+\frac{3}{2}x^2-3x^3\right)}{x(2x-2x^2)}\right] = \lim_{x\to 0}\left[\frac{-\frac{3}{2}x^2+3x^3}{2x^2-2x^3}\right] = \lim_{x\to 0}\left[\frac{-\frac{3}{2}+3x}{2-2x}\right] = \frac{-\frac{3}{2}}{2} = -\frac{3}{4}$

8) a)

$\ln(1+3x) = 3x - \frac{(3x)^2}{2} + \frac{(3x)^3}{3} - \frac{(3x)^4}{4} + \cdots \approx 3x - \frac{9}{2}x^2 + 9x^3 - \frac{81}{4}x^4$

8) b)

$\ln(1-3x) \approx -3x - \frac{9}{2}x^2 - 9x^3 - \frac{81}{4}x^4$

So $\ln[(1+3x)(1-3x)] = \ln(1+3x) + \ln(1-3x)$

$\approx \left(3x - \frac{9}{2}x^2 + 9x^3 - \frac{81}{4}x^4\right) + \left(-3x - \frac{9}{2}x^2 - 9x^3 - \frac{81}{4}x^4\right)$

$= -9x^2 - \frac{81}{2}x^4$

8) c)

$$\text{So } \lim_{x\to 0}\left[\frac{\ln[(1+3x)(1-3x)]}{4x^2}\right] \approx \lim_{x\to 0}\left[\frac{-9x^2-\frac{81}{2}x^4}{4x^2}\right] = \lim_{x\to 0}\left[-\frac{9}{4}-\frac{81}{8}x^2\right] = -\frac{9}{4}$$

9) a)

$$e^{5x^2} = 1 + 5x^2 + \frac{1}{2}(5x^2)^2 + \cdots \approx 1 + 5x^2 + \frac{25}{2}x^4$$

9) b)

$$\text{So } \lim_{x\to 0}\left[\frac{e^{5x^2}}{k-\cos x}\right] \approx \lim_{x\to 0}\left[\frac{1+5x^2+\frac{25}{2}x^4}{k-\left(1-\frac{1}{2}x^2+\frac{1}{24}x^4\right)}\right] = \frac{1}{k-1} = -6$$

$$\therefore k = \frac{5}{6}$$

10) a)

$$\frac{6(1-x)}{(x+1)(x+3)} \equiv \frac{A}{x+1} + \frac{B}{x+3}$$

$$6 - 6x = A(x+3) + B(x+1)$$

Let $x = -1$: $\quad 12 = 2A \therefore A = 6$

Let $x = -3$: $\quad 24 = -2B \therefore B = -12$

$$\frac{6(1-x)}{(x+1)(x+3)} = \frac{6}{x+1} - \frac{12}{x+3}$$

10) b)

$$f(x) = \frac{6}{x+1} - \frac{12}{x+3} = 6(1+x)^{-1} - 12(3+x)^{-1}$$

$$= 6(1+x)^{-1} - 4\left(1+\frac{1}{3}x\right)^{-1}$$

$$\approx 6[1 - x + x^2] - 4\left[1 - \frac{1}{3}x + \frac{1}{9}x^2\right]$$

$$= 6 - 6x + 6x^2 - 4 + \frac{4}{3}x - \frac{4}{9}x^2$$

$$= 2 - \frac{14}{3}x + \frac{50}{9}x^2$$

10) c)

$$\lim_{x\to 0}[f(x)] \approx \lim_{x\to 0}\left[2 - \frac{14}{3}x + \frac{50}{9}x^2\right] = 2$$

11) a)

$\frac{3-x}{(x+1)(x-5)} \equiv \frac{A}{x+1} + \frac{B}{x-5}$

$3 - x = A(x-5) + B(x+1)$

Let $x = -1$: $\quad 4 = -6A \therefore A = -\frac{2}{3}$

Let $x = 5$: $\quad -2 = 6B \therefore B = -\frac{1}{3}$

$\frac{3-x}{(x+1)(x-5)} = -\frac{2}{3(x+1)} - \frac{1}{3(x-5)}$

11) b)

$f(x) = -\frac{2}{3(x+1)} - \frac{1}{3(x-5)}$

$= -\frac{2}{3}(x+1)^{-1} - \frac{1}{3}(x-5)^{-1}$

$\approx -\frac{2}{3}[1 - x + x^2] - \frac{1}{3}\left[-\frac{1}{5} - \frac{1}{25}x - \frac{1}{125}x^2\right]$

$= -\frac{2}{3} + \frac{2}{3}x - \frac{2}{3}x^2 + \frac{1}{15} + \frac{1}{75}x + \frac{1}{375}x^2$

$= -\frac{3}{5} + \frac{17}{25}x - \frac{83}{125}x^2$

11) c)

$\lim_{x\to 0}\left[\frac{1}{2}f(x)\right] \approx \lim_{x\to 0}\left[\frac{3-x}{2(x+1)(x-5)}\right] = \lim_{x\to 0}\left[\frac{1}{2}\left(-\frac{3}{5} + \frac{17}{25}x - \frac{83}{125}x^2\right)\right] = -\frac{3}{10}$

12) a)

$\frac{4}{(x-1)(x^2+2)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+2}$

$4 = A(x^2+2) + (Bx+C)(x-1)$

Let $x = 1$: $\quad 4 = 3A \therefore A = \frac{4}{3}$

Let $x = 0$: $\quad 4 = 2A - C \therefore C = -\frac{4}{3}$

Let $x = -1$: $\quad 4 = 3A + 2B - 2C \therefore B = -\frac{4}{3}$

$\frac{4}{(x-1)(x^2+2)} = \frac{4}{3(x-1)} - \frac{4x+4}{3(x^2+2)}$

12) b)

$$f(x) = \frac{4}{3(x-1)} - \frac{4x+4}{3(x^2+2)} = \frac{4}{3}(-1+x)^{-1} - \frac{4}{3}(x+1)(2+x^2)^{-1}$$

$$= -\frac{4}{3}(1-x)^{-1} - \frac{2}{3}(x+1)\left(1+\frac{x^2}{2}\right)^{-1}$$

$$\approx -\frac{4}{3}[1+x+x^2] - \frac{2}{3}(x+1)\left[1-\frac{1}{2}x^2\right]$$

$$= -\frac{4}{3} - \frac{4}{3}x - \frac{4}{3}x^2 - \frac{2}{3} - \frac{2}{3}x + \frac{1}{3}x^2 + \cdots$$

$$= -2 - 2x - x^2$$

12) c)

$$\lim_{x\to 0}[f(x)] \approx \lim_{x\to 0}[-2-2x-x^2] = -2$$

13) a)

$$\cos 2x = 1 - \frac{1}{2}(2x)^2 + \frac{1}{24}(2x)^4 - \frac{1}{720}(2x)^6 \ldots \approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6$$

13) b)

$$\text{So } \lim_{x\to 0}\left[\frac{1-\cos 2x}{x\ln\left(1+\frac{1}{2}x\right)}\right] \approx \lim_{x\to 0}\left[\frac{1-\left(1-2x^2+\frac{2}{3}x^4-\frac{4}{45}x^6\right)}{x\ln\left(1+\frac{1}{2}x\right)}\right]$$

$$= \lim_{x\to 0}\left[\frac{2x^2-\frac{2}{3}x^4+\frac{4}{45}x^6}{x\ln\left(1+\frac{1}{2}x\right)}\right] = \lim_{x\to 0}\left[\frac{2x-\frac{2}{3}x^3+\frac{4}{45}x^5}{\ln\left(1+\frac{1}{2}x\right)}\right]$$

Using L'Hôpital's rule:

$$\lim_{x\to 0}\left[\frac{2x-\frac{2}{3}x^3+\frac{4}{45}x^5}{\ln\left(1+\frac{1}{2}x\right)}\right] = \lim_{x\to 0}\left[\frac{2-2x^2+\frac{4}{9}x^4}{\frac{1}{2\left(1+\frac{1}{2}x\right)}}\right] = \frac{2}{\frac{1}{2}} = 4$$

Alternatively:

$$\text{So } \lim_{x\to 0}\left[\frac{1-\cos 2x}{x\ln\left(1+\frac{1}{2}x\right)}\right] \approx \lim_{x\to 0}\left[\frac{1-\left(1-2x^2+\frac{2}{3}x^4-\frac{4}{45}x^6\right)}{x\times\left(\frac{x}{2}-\frac{x^2}{8}+\cdots\right)}\right]$$

$$= \lim_{x\to 0}\left[\frac{1-\left(1-2x^2+\frac{2}{3}x^4-\frac{4}{45}x^6\right)}{\frac{x^2}{2}-\frac{x^3}{8}}\right] = \lim_{x\to 0}\left[\frac{2x^2-\frac{2}{3}x^4+\frac{4}{45}x^6}{\frac{x^2}{2}-\frac{x^3}{8}}\right]$$

Using L'Hôpital's rule:

$$\lim_{x\to 0}\left[\frac{2x^2-\frac{2}{3}x^4+\frac{4}{45}x^6}{\frac{x^2}{2}-\frac{x^3}{8}}\right] = \lim_{x\to 0}\left[\frac{4x-\frac{8}{3}x^3+\frac{8}{15}x^5}{x-\frac{3x^2}{8}}\right] = \lim_{x\to 0}\left[\frac{4-8x^2+\frac{8}{3}x^4}{1-\frac{3x}{4}}\right] = \frac{4}{1} = 4$$

14) a)

$e^{-3x} = 1 - 3x + \frac{1}{2}(-3x)^2 + \frac{1}{6}(-3x)^3 + \cdots \approx 1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3$

14) b)

So $\lim_{x\to 0}\left[\frac{1-e^{-3x}}{\ln(1+kx)}\right] \approx \lim_{x\to 0}\left[\frac{1-\left(1-3x+\frac{9}{2}x^2-\frac{9}{2}x^3+\cdots\right)}{\ln(1+kx)}\right] = \lim_{x\to 0}\left[\frac{3x-\frac{9}{2}x^2+\frac{9}{2}x^3}{\ln(1+kx)}\right]$

Using L'Hôpital's rule:

$\lim_{x\to 0}\left[\frac{3x-\frac{9}{2}x^2+\frac{9}{2}x^3}{\ln(1+kx)}\right] = \lim_{x\to 0}\left[\frac{3-9x+\frac{27}{2}x^2}{\frac{k}{1+kx}}\right]$

So that

$\frac{3}{k} = \frac{1}{2} \therefore k = 6$

Alternatively:

So $\lim_{x\to 0}\left[\frac{1-e^{-3x}}{\ln(1+kx)}\right] \approx \lim_{x\to 0}\left[\frac{1-\left(1-3x+\frac{9}{2}x^2-\frac{9}{2}x^3+\cdots\right)}{kx-\frac{k^2}{2}x^2+\cdots}\right] = \lim_{x\to 0}\left[\frac{3x-\frac{9}{2}x^2+\frac{9}{2}x^3}{kx-\frac{k^2}{2}x^2+\cdots}\right]$

Using L'Hôpital's rule:

$\lim_{x\to 0}\left[\frac{3x-\frac{9}{2}x^2+\frac{9}{2}x^3}{kx-\frac{k^2}{2}x^2+\cdots}\right] = \lim_{x\to 0}\left[\frac{3-9x+\frac{27}{2}x^2}{k-k^2x+\cdots}\right]$

So that

So that

$\frac{3}{k} = \frac{1}{2} \therefore k = 6$

15) a)

$\ln(e^{2x}\sin x) = \ln(e^{2x}) + \ln(\sin x) = 2x + \ln(\sin x)$

$\sin x = x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \cdots$

So $\ln(\sin x) = \ln\left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \cdots\right) = \ln\left(x\left(1 - \frac{1}{6}x^2 + \frac{1}{120}x^4 - \cdots\right)\right)$

$= \ln x + \ln\left(1 - \frac{1}{6}x^2 + \frac{1}{120}x^4 - \cdots\right)$

$=\ln x + \left(-\frac{1}{6}x^2 + \frac{1}{120}x^4\right) - \frac{1}{2}\left(-\frac{1}{6}x^2 + \frac{1}{120}x^4\right)^2 + \cdots$

$\therefore \ln(e^{2x}\sin x) \approx \ln x + 2x - \frac{1}{6}x^2$ as req.

15) b)

$\ln(1 + kx) = kx - \frac{1}{2}(kx)^2 + \cdots \approx kx - \frac{k^2}{2}x^2$

15) c)

So $\lim_{x\to 0}\left[\frac{\ln(e^{2x}\sin x)-\ln x}{\ln(1+kx)}\right] \approx \lim_{x\to 0}\left[\frac{\ln x+2x-\frac{1}{6}x^2-\ln x}{kx-\frac{k^2}{2}x^2}\right] = \lim_{x\to 0}\left[\frac{2x-\frac{1}{6}x^2}{kx-\frac{k^2}{2}x^2}\right] = \lim_{x\to 0}\left[\frac{2-\frac{1}{6}x}{k-\frac{k^2}{2}x}\right] = \frac{2}{k}$

So it exists for any $k \neq 0$ and $\lim_{x\to 0}\left[\frac{\ln(e^{2x}\sin x)-\ln x}{\ln(1+kx)}\right] = \frac{2}{k}$

16) a)

$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$

16) b)

$$\sum_{r=1}^{\infty}\frac{4-r}{r!} = 4\sum_{r=1}^{\infty}\frac{1}{r!} - \sum_{r=1}^{\infty}\frac{r}{r!}$$

Since $e^x - 1 = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$

$$\sum_{r=1}^{\infty}\frac{1}{r!} = e^1 - 1$$

and since $\frac{r}{r!} = \frac{1}{(r-1)!}$

$$\sum_{r=1}^{\infty}\frac{r}{r!} = \sum_{r=1}^{\infty}\frac{1}{(r-1)!} = e^1$$

$$\therefore \sum_{r=1}^{\infty}\frac{4-r}{r!} = 4(e-1) - e = 3e - 4$$

16) c)

Since $e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!}$ then $e^x - 1 = \sum_{r=1}^{\infty} \frac{x^r}{r!}$

So that

$$4(e^x - 1) - xe^x = 4\left(\sum_{r=1}^{\infty} \frac{x^r}{r!}\right) - x\left(\sum_{r=0}^{\infty} \frac{x^r}{r!}\right) = \sum_{r=1}^{\infty} \frac{4x^r}{r!} - \sum_{r=0}^{\infty} \frac{x^{r+1}}{r!}$$

$$= \sum_{r=1}^{\infty} \frac{4x^r}{r!} - \sum_{r=1}^{\infty} \frac{x^r}{(r-1)!}$$

$$= \sum_{r=1}^{\infty} \left(\frac{4x^r}{r!} - \frac{x^r}{(r-1)!}\right) = \sum_{r=1}^{\infty} \left(\frac{4}{r!} - \frac{1}{(r-1)!}\right) x^r = \sum_{r=1}^{\infty} \left(\frac{4}{r!} - \frac{r}{r!}\right) x^r = \sum_{r=1}^{\infty} \left(\frac{4-r}{r!}\right) x^r$$

$$\therefore \frac{4(e^x - 1) - xe^x}{x} = \sum_{r=1}^{\infty} \left(\frac{4-r}{r!}\right) x^{r-1}$$

$$\lim_{x \to 0} \left[\frac{4(e^x - 1) - xe^x}{x}\right] = \lim_{x \to 0} \left[\sum_{r=1}^{\infty} \left(\frac{4-r}{r!}\right) x^{r-1}\right] = \frac{4-1}{1!} = 3$$

17) a)

$f(x) = \sqrt{4-x}$ so $f(0) = 2$

$f'(x) = -\frac{1}{2}(4-x)^{-\frac{1}{2}}$ so $f'(0) = -\frac{1}{4}$

$f''(x) = -\frac{1}{4}(4-x)^{-\frac{3}{2}}$ so $f''(0) = -\frac{1}{32}$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$$

So $\sqrt{4-x} = 2 - \frac{1}{4}x - \frac{1}{64}x^2 + \cdots$

$\therefore \sqrt{4-x} \approx 2 - \frac{1}{4}x - \frac{1}{64}x^2$

17) b)

So $\lim_{x \to 0} \left[\frac{\sqrt{4-x}}{1+x}\right] \approx \lim_{x \to 0} \left[\frac{2 - \frac{1}{4}x - \frac{1}{64}x^2}{1+x}\right] = 2$

18) a)

$\mathrm{f}(x) = \sqrt{2 + \mathrm{e}^x}$ so $\mathrm{f}(0) = \sqrt{3}$

$\mathrm{f}'(x) = \frac{1}{2}\mathrm{e}^x(2 + \mathrm{e}^x)^{-\frac{1}{2}}$ so $\mathrm{f}'(0) = \frac{1}{2\sqrt{3}}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sqrt{2 + \mathrm{e}^x} = \sqrt{3} + \frac{\sqrt{3}}{6}x + \cdots$

$\therefore \sqrt{2 + \mathrm{e}^x} \approx \sqrt{3} + \frac{\sqrt{3}}{6}x$

18) b)

So $\lim_{x\to 0}\left[\frac{\sqrt{2+\mathrm{e}^x}-1}{\cos 2x}\right] \approx \lim_{x\to 0}\left[\frac{\left(\sqrt{3}+\frac{\sqrt{3}}{6}x\right)-1}{1-2x^2}\right] = \sqrt{3} - 1$

19) a)

$\mathrm{f}(x) = \sqrt{1 + \tan x}$ so $\mathrm{f}(0) = 1$

$\mathrm{f}'(x) = \frac{1}{2}\sec^2 x\,(1 + \tan x)^{-\frac{1}{2}}$ so $\mathrm{f}'(0) = \frac{1}{2}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sqrt{1 + \tan x} = 1 + \frac{1}{2}x + \cdots$

$\therefore \sqrt{1 + \tan x} \approx 1 + \frac{1}{2}x$

19) b)

So $\lim_{x\to 0}\left[\frac{\sqrt{1+\tan x}-1}{\sin kx}\right] \approx \lim_{x\to 0}\left[\frac{\left(1+\frac{1}{2}x\right)-1}{kx-\frac{k}{6}x^3}\right] = \lim_{x\to 0}\left[\frac{\frac{1}{2}x}{kx-\frac{k}{6}x^3}\right] = \lim_{x\to 0}\left[\frac{\frac{1}{2}}{k-\frac{k}{6}x^2}\right] = \frac{\frac{1}{2}}{k} = \frac{1}{4}$

$\therefore k = 2$

20) a)

$f(x) = (1+\sin x)^{\frac{3}{2}}$ so $f(0) = 1$

$f'(x) = \frac{3}{2}\cos x\,(1+\sin x)^{\frac{1}{2}}$ so $f'(0) = \frac{3}{2}$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $(1+\sin x)^{\frac{3}{2}} = 1 + \frac{3}{2}x + \cdots$

$\therefore (1+\sin x)^{\frac{3}{2}} \approx 1 + \frac{3}{2}x$

20) b)

So $\lim_{x\to 0}\left[\frac{(1+\sin x)^{\frac{3}{2}}-1}{\sin kx}\right] \approx \lim_{x\to 0}\left[\frac{\left(1+\frac{3}{2}x\right)-1}{kx-\frac{k^3}{6}x^3}\right] = \lim_{x\to 0}\left[\frac{\frac{3}{2}x}{kx-\frac{k^3}{6}x^3}\right] = \lim_{x\to 0}\left[\frac{\frac{3}{2}}{k-\frac{k^3}{6}x^2}\right] = \frac{\frac{3}{2}}{k} = \frac{3}{8}$

$\therefore k = 4$

21) a)

$f(x) = \cosh x$ so $f(0) = 1$

$f'(x) = \sinh x$ so $f'(0) = 0$

$f''(x) = \cosh x$ so $f''(0) = 1$

$f^{(3)}(x) = \sinh x$ so $f^{(3)}(0) = 0$

$f^{(4)}(x) = \cosh x$ so $f^{(4)}(0) = 1$

$f^{(5)}(x) = \sinh x$ so $f^{(3)}(0) = 0$

$f^{(6)}(x) = \cosh x$ so $f^{(4)}(0) = 1$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\cosh x = 1 + 0 + \frac{1}{2!}x^2 + 0 + \frac{1}{4!}x^4 + 0 + \frac{1}{6!}x^6 \ldots$

$\therefore \cosh x \approx 1 + \frac{1}{2}x^2 + \frac{1}{24}x^4$

21) b)

So $\lim_{x\to 0}\left[\frac{\cosh x - \cos x}{x^2}\right] = \lim_{x\to 0}\left[\frac{1+\frac{1}{2}x^2+\frac{1}{24}x^4+\cdots-\left(1-\frac{1}{2}x^2+\frac{1}{24}x^4-\cdots\right)}{x^2}\right] \approx \lim_{x\to 0}\left[\frac{x^2}{x^2}\right] = 1$

22) a)

$(1+y)^{-1} \approx 1 - y + y^2$

22) b)

$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{1}{1+(\cosh x - 1)} \approx \frac{1}{1+\left(\frac{1}{2}x^2+\frac{1}{24}x^4-1\right)}$

$\approx 1 - \left(\frac{1}{2}x^2 + \frac{1}{24}x^4\right) + \left(\frac{1}{2}x^2 + \frac{1}{24}x^4\right)^2$

$\approx 1 - \frac{1}{2}x^2 - \frac{1}{24}x^4 + \frac{1}{4}x^4 + \cdots$

$\approx 1 - \frac{1}{2}x^2 + \frac{5}{24}x^4$ as req.

22) c)

$\mathrm{f}(x) = \sinh x$ so $\mathrm{f}(0) = 0$

$\mathrm{f}'(x) = \cosh x$ so $\mathrm{f}'(0) = 1$

$\mathrm{f}''(x) = \sinh x$ so $\mathrm{f}''(0) = 0$

$\mathrm{f}^{(3)}(x) = \cosh x$ so $\mathrm{f}^{(3)}(0) = 1$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sinh x = 0 + x + 0 + \frac{1}{3!}x^3 + \cdots$

$\therefore \sinh x \approx x + \frac{1}{6}x^3$ as req.

22) d)

So $\lim_{x\to 0}\left[\frac{x\sinh 3x}{\operatorname{sech}x - 1}\right] \approx \lim_{x\to 0}\left[\frac{x\left(3x+\frac{9}{2}x^3\right)}{\left(1-\frac{1}{2}x^2+\frac{5}{24}x^4\right)-1}\right] = \lim_{x\to 0}\left[\frac{3x^2+\frac{9}{2}x^4}{-\frac{1}{2}x^2+\frac{5}{24}x^4}\right] = \lim_{x\to 0}\left[\frac{3+\frac{9}{2}x^2}{-\frac{1}{2}+\frac{5}{24}x^2}\right] = \frac{3}{-\frac{1}{2}} = -6$

23) a)

$\mathrm{f}(x) = \mathrm{e}^{\sin x}$ so $\mathrm{f}(0) = 1$

$\mathrm{f}'(x) = \cos x\,\mathrm{e}^{\sin x}$ so $\mathrm{f}'(0) = 1$

$\mathrm{f}''(x) = \cos^2 x\,\mathrm{e}^{\sin x} - \sin x\,\mathrm{e}^{\sin x}$ so $\mathrm{f}''(0) = 1$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\mathrm{e}^{\sin x} = 1 + x + \frac{1}{2!}x^2 + \cdots$

$\therefore \mathrm{e}^{\sin x} \approx 1 + x + \frac{1}{2}x^2$

23) b)

So $\lim_{x\to 0}\left[\frac{\mathrm{e}^{\sin x}-1}{2x\cos x}\right] \approx \lim_{x\to 0}\left[\frac{\left(1+x+\frac{1}{2}x^2\right)-1}{2x\left(1-\frac{1}{2}x^2+\cdots\right)}\right] = \lim_{x\to 0}\left[\frac{x+\frac{1}{2}x^2}{2x-x^3}\right] = \lim_{x\to 0}\left[\frac{1+\frac{1}{2}x}{2-x^2}\right] = \frac{1}{2}$

24) a)

$\ln(1+2x) = 2x - \frac{1}{2}(2x)^2 + \cdots \approx 2x - 2x^2$

24) b)

$f(x) = \ln(\cos x)$ so $f(0) = 0$

$f'(x) = -\frac{\sin x}{\cos x}$ so $f'(0) = 0$

$f''(x) = \frac{-\cos^2 x - \sin^2 x}{\cos^2 x} = -\frac{1}{\cos^2 x}$ so $f''(0) = -1$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\ln(\cos x) = 0 + 0 - \frac{1}{2!}x^2 + \cdots$

$\therefore \ln(\cos x) \approx -\frac{1}{2}x^2$

24) c)

So $\lim_{x\to 0}\left[\frac{x\ln(1+2x)}{\ln(\cos x)-x^2}\right] \approx \lim_{x\to 0}\left[\frac{x(2x-2x^2)}{-\frac{1}{2}x^2-x^2}\right] = \lim_{x\to 0}\left[\frac{2x^2-2x^3}{-\frac{3}{2}x^2}\right] = \lim_{x\to 0}\left[\frac{2-2x}{-\frac{3}{2}}\right] = \frac{2}{-\frac{3}{2}} = -\frac{4}{3}$

25) a)

$f(x) = \ln(1+2\sin x)$ so $f(0) = 0$

$f'(x) = \frac{2\cos x}{1+2\sin x}$ so $f'(0) = 2$

$f''(x) = \frac{-2\sin x(1+2\sin x) - 4\cos^2 x}{(1+2\sin x)^2}$ so $f''(0) = -4$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\ln(1+2\sin x) = 0 + 2x - \frac{4}{2!}x^2 + \cdots$

$\therefore \ln(1+2\sin x) \approx 2x - 2x^2$

25) b)

So $\lim_{x\to 0}\left[\frac{\ln(1+2\sin x)}{\ln(1-x)}\right] \approx \lim_{x\to 0}\left[\frac{2x-2x^2}{-x-\frac{1}{2}x^2}\right] = \lim_{x\to 0}\left[\frac{2-2x}{-1-\frac{1}{2}x}\right] = -2$

26) a)

$\mathrm{f}(x) = \ln(2 - \mathrm{e}^x)$ so $\mathrm{f}(0) = 0$

$\mathrm{f}'(x) = -\frac{\mathrm{e}^x}{2-\mathrm{e}^x}$ so $\mathrm{f}'(0) = -1$

$\mathrm{f}''(x) = \frac{-\mathrm{e}^x(2-\mathrm{e}^x)-\mathrm{e}^{2x}}{(2-\mathrm{e}^x)^2}$ so $\mathrm{f}''(0) = -2$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\ln(2 - \mathrm{e}^x) = 0 - x - \frac{2}{2!}x^2 + \cdots$

$\therefore \ln(2 - \mathrm{e}^x) \approx -x - x^2$

26) b)

So $\lim_{x\to 0}\left[\frac{x\ln(2-\mathrm{e}^x)}{1-\cos kx}\right] \approx \lim_{x\to 0}\left[\frac{x(-x-x^2)}{1-\left(1-\frac{k^2}{2}x^2\right)}\right] = \lim_{x\to 0}\left[\frac{-x^2-x^3}{\frac{k^2}{2}x^2}\right] = \lim_{x\to 0}\left[-\frac{2}{k^2} - \frac{2}{k^2}x\right] = -\frac{2}{k^2}$

So $-\frac{2}{k^2} = -\frac{1}{8}$ gives $k = \pm 4$

27)

Substitute $x = \frac{\pi}{4} + t$:

$\frac{\mathrm{d}x}{\mathrm{d}t} = 1$, when $x = \frac{\pi}{4}$, $t = 0$ and when $x = \frac{\pi}{4} + h$, $t = h$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{4}+h} (\cos x)^{\frac{1}{3}}\mathrm{d}x = \int_0^h \left(\cos\left(\frac{\pi}{4}+t\right)\right)^{\frac{1}{3}}\mathrm{d}t$$

$\mathrm{f}(t) = \left(\cos\left(\frac{\pi}{4}+t\right)\right)^{\frac{1}{3}}$ so $\mathrm{f}(0) = \left(\frac{\sqrt{2}}{2}\right)^{\frac{1}{3}}$

So $\left(\cos\left(\frac{\pi}{4}+t\right)\right)^{\frac{1}{3}} = \left(\frac{\sqrt{2}}{2}\right)^{\frac{1}{3}} + \cdots$

Since we will be taking the limit $h \to 0$, then $\left(\cos\left(\frac{\pi}{4}+t\right)\right)^{\frac{1}{3}} \approx \left(\frac{\sqrt{2}}{2}\right)^{\frac{1}{3}}$

So

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{4}+h} \frac{3\sqrt[3]{\cos x}}{2h}\,\mathrm{d}x = \int_0^h \frac{3\left(\cos\left(\frac{\pi}{4}+t\right)\right)^{\frac{1}{3}}}{2h}\,\mathrm{d}t \approx \int_0^h \frac{3}{2h} \times \left(\frac{\sqrt{2}}{2}\right)^{\frac{1}{3}}\mathrm{d}t = \left[\frac{3}{2h}\left(\frac{\sqrt{2}}{2}\right)^{\frac{1}{3}}t\right]_0^h = 3 \times 2^{-\frac{7}{6}}$$

$$\therefore \lim_{h\to 0}\int_{\frac{\pi}{4}}^{\frac{\pi}{4}+h} \frac{3\sqrt[3]{\cos x}}{2h}\,\mathrm{d}x = \frac{3}{2^{\frac{7}{6}}}$$